CRYONANO

AC-DC Converter Systems

Precision Power Conversion for Industrial, Defense & Mission-Critical Applications

Reliable Power Conversion.
Stable High-Power DC Output.
Built for Advanced Applications.

CRYONANO Intelligent Battery Chargers are high-reliability power systems engineered to deliver safe, stable, and efficient charging across a wide range of battery chemistries and operating environments.
Designed for embedded systems, off-grid installations, and mission-critical platforms, these chargers provide precise charging control, advanced protection, and long-term operational stability. From laboratory setups to defense and aerospace deployments, CRYONANO chargers ensure uninterrupted power where uptime and durability are essential.

Notes

- System performance depends on load conditions and environmental factors
- Custom solutions available for defense, aerospace, and harsh environments
- Compliance and certification options available upon request
- Thermal management and cooling vary by configuration

KEY FEATURES

- **High-Efficiency Power Conversion**
  Up to 95% conversion efficiency minimizes thermal losses while delivering high power density and improved system reliability.
- **Wide Input Compatibility**
  Supports 85–265 VAC, 47–63 Hz input range with compatibility across single-phase and three-phase systems for global deployment.
- **Integrated Protection System**
  Includes over-voltage, over-current, short-circuit, and over-temperature protection for reliable operation in harsh environments.
- **Active Power Factor Correction (PFC)**
  Integrated active PFC improves input power quality, reduces harmonic distortion, and meets industrial efficiency standards.

APPLICATIONS

- **Industrial automation and PLC systems**
- **Test and measurement equipment**
- **Laboratory and scientific instrumentation**
- **Embedded electronics and control systems**
- **Telecom and networking infrastructure**
- **Battery charging systems and power front-ends**

AC-DC Converter Systems

SPECIFICATIONS

ELECTRICAL PERFORMANCE

Input Voltage Range	85–265 VAC
Input Frequency	47–63 Hz
Output Voltage	12–48 V DC (configurable)
Efficiency	Up to 95%
Power Range	300 W to 60 kW

PROTECTION AND COMPLIANCE

Over-Voltage Protection	Integrated
Over-Current Protection	Integrated
Short-Circuit Protection	Integrated
Over-Temperature Protection	Integrated
Power Factor Correction	Active PFC

MECHANICAL AND SYSTEM DESIGN

Form Factor	Compact to rack-integrated systems
Cooling	Passive / forced-air (configuration dependent)
Power Density	High-density design
Build Type	Industrial / ruggedized

CONFIGURABILITY

Architecture Options	COTS / Modified / Custom
Output Configuration	Single / multiple outputs
Integration	Embedded or standalone systems
Customization	Voltage, form factor, environment-specific

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